Linux Shell Basic Commands Handout

Introduction

The Linux shell is a text-based interface for interacting with a Linux-based operating system. Mastering basic shell commands is essential for various IT tasks, including system administration, software development, and troubleshooting.

Accessing the Linux Shell

Terminal Emulators

- Terminal emulators are applications that provide access to the Linux shell.
- Common terminal emulators include GNOME Terminal, Konsole, xterm, and iTerm (for macOS).

Basic Linux Commands

1. `ls` - List Files and Directories

- List files and directories in the current location.
- Example: `ls`

2. `cd` - Change Directory

- Change the current working directory.
- Example: `cd /path/to/directory`

3. `pwd` - Print Working Directory

- Display the current working directory.
- Example: `pwd`

4. `mkdir` - Create Directory

- Create a new directory.
- Example: `mkdir my_directory`

5. `touch` - Create Empty File

- Create an empty file.
- Example: `touch myfile.txt`

6. `cp` - Copy Files and Directories

- Copy files and directories.
- Example: `cp file1.txt file2.txt`

7. `mv` - Move or Rename Files and Directories

- Move or rename files and directories.
- Example: `mv oldfile.txt newfile.txt`

8. `rm` - Remove Files and Directories

- Delete files and directories.
- Example: `rm unwanted_file.txt`

9. `cat` - View File Contents

- Display the contents of a text file.
- Example: `cat myfile.txt`

10. `man` - Access Manual Pages

- Access the manual pages for commands.
- Example: `man ls`

Activities for Practice

Activity 1: Navigation and File Manipulation

- Navigate to different directories using `cd`.
- Create, copy, move, and delete files and directories.

Activity 2: Redirection and Pipes

- Use `>` to redirect output to a file and `>>` to append output to a file.
- Practice piping (`|`) to combine the output of one command as input to another.

Activity 3: Users and Permissions

- Learn about users and groups using `whoami`, `id`, and `groups`.
- Experiment with changing file permissions using `chmod`, `chown`, and `chgrp`.

Conclusion

- Mastering Linux Shell Basic Commands is crucial for IT professionals.
- Continue to practice and explore more advanced shell commands.
- Use the Linux shell effectively in your IT tasks and projects.